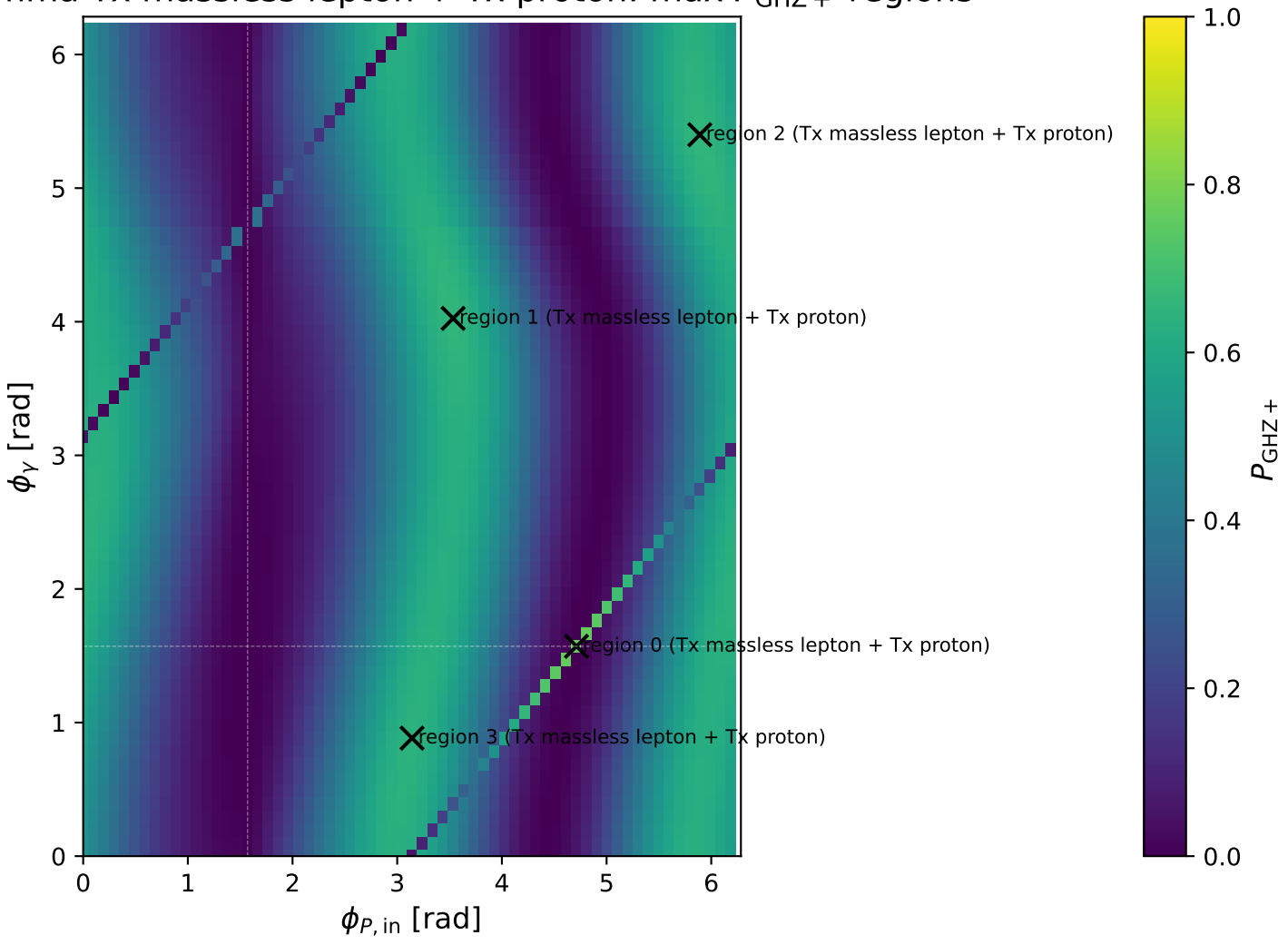
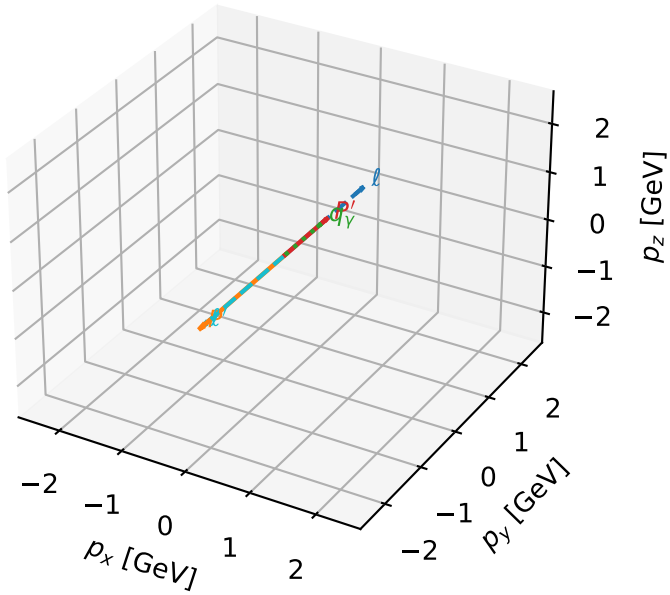


mma Tx massless lepton + Tx proton: max $P_{\text{GHZ}+}$ regions



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta

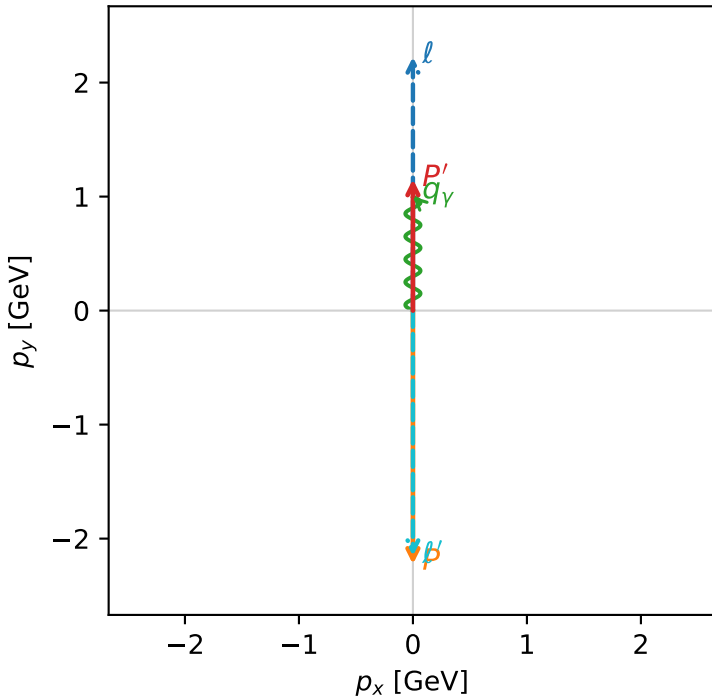


ghz_purity_mid_egamma_txtx_region_0 $P_{\{\text{rm GHZ}\}}=0.767466$
 region: mid_Egamma TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

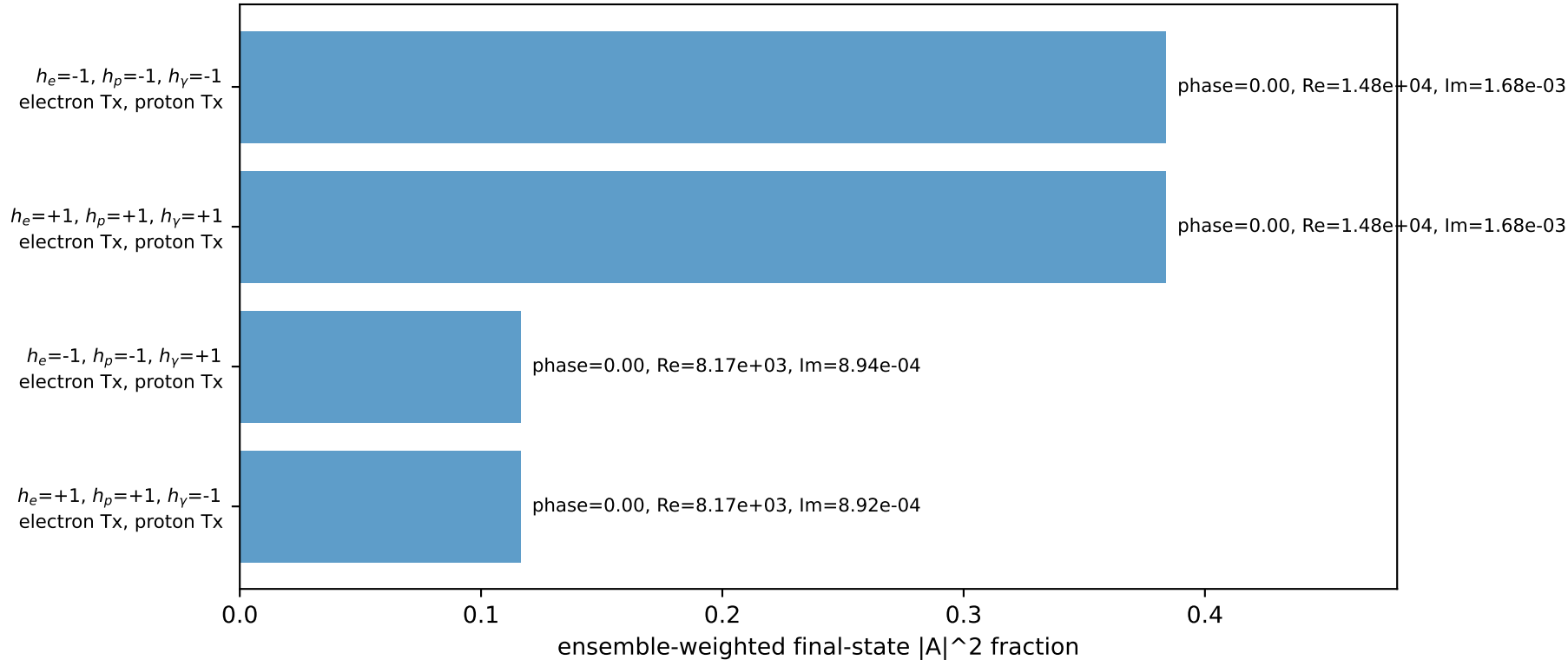
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.15253$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_p=4.71239$
 $E_{\gamma}=1$, $\phi_{\gamma}=1.5708$
 $Q^2=19.1525$, $x_B=0.96635$, $t=-10.5424$, $W^2=1.54677$, $y=0.960429$
 $\theta(\ell', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 2.1525$ $p_x= -0.0000$ $p_y= -2.1525$ $p_z= 0.0000$
 P' $E= 1.4860$ $p_x= 0.0000$ $p_y= 1.1525$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.0000$ $p_y= 1.0000$ $p_z= 0.0000$

Transverse plane

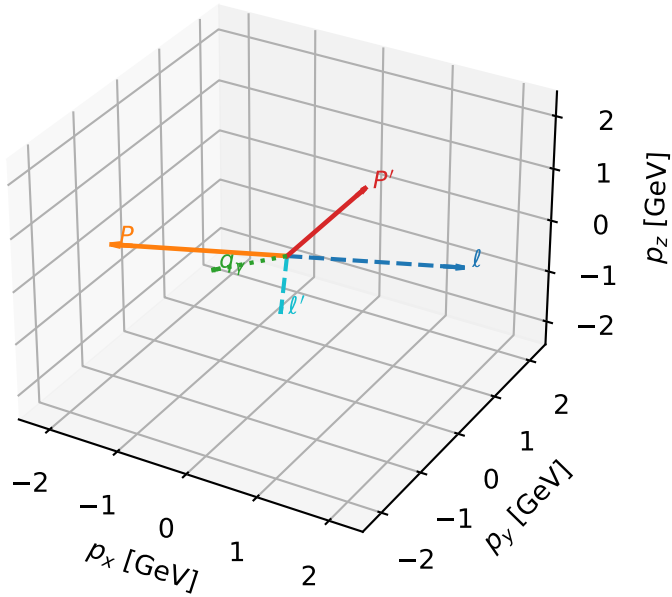


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta

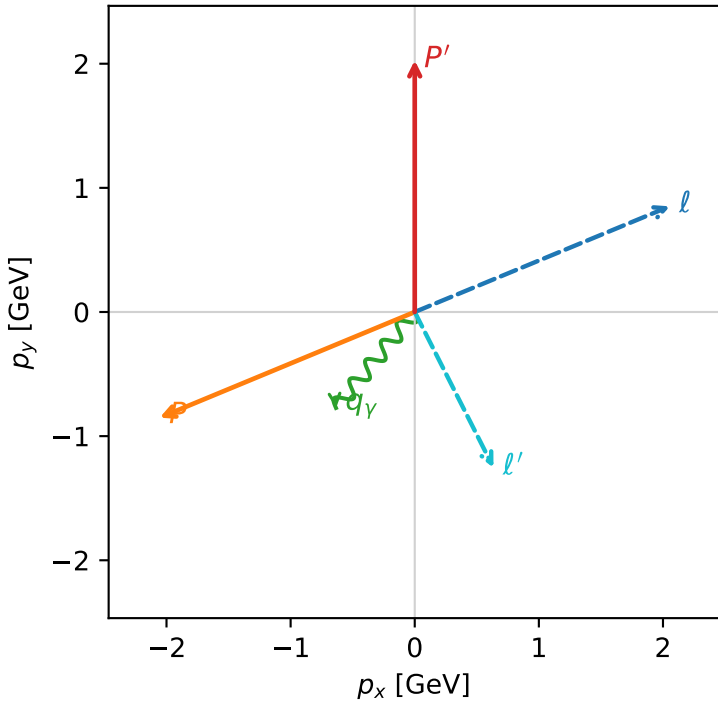


ghz_purity_mid_egamma_txtx_region_1 $P_{\text{GHZ}}=0.65415$
 region: mid_Egamma_TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

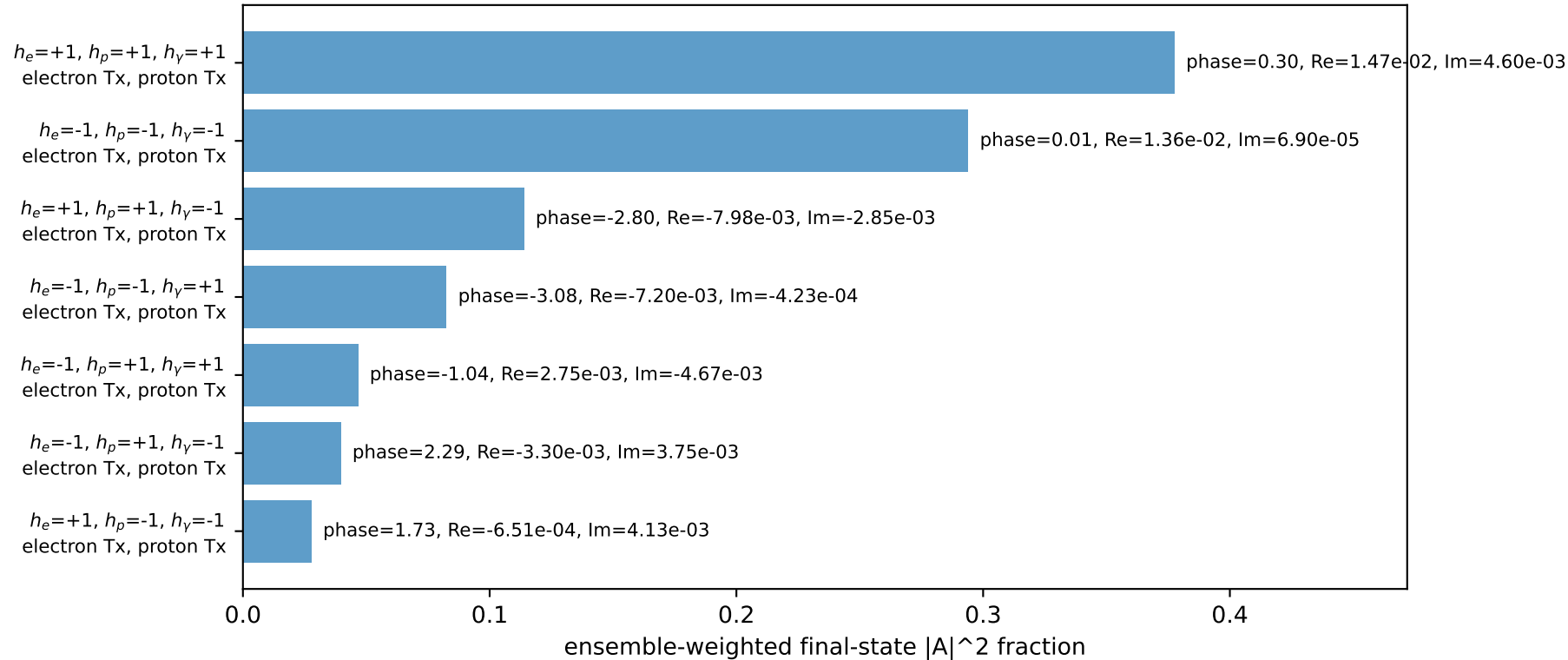
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.02682$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=0.392699$, $\phi_p=3.53429$
 $E_\gamma=1$, $\phi_\gamma=4.02517$
 $Q^2=5.77849$, $x_B=0.431917$, $t=-12.474$, $W^2=8.48005$, $y=0.648319$
 $\theta(\ell', \gamma)=66.2131$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 2.0551$ $p_y= 0.8512$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -2.0551$ $p_y= -0.8512$ $p_z= 0.0000$
 ℓ' $E= 1.4052$ $p_x= 0.6344$ $p_y= -1.2538$ $p_z= 0.0000$
 P' $E= 2.2333$ $p_x= 0.0000$ $p_y= 2.0268$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.6344$ $p_y= -0.7730$ $p_z= 0.0000$

Transverse plane

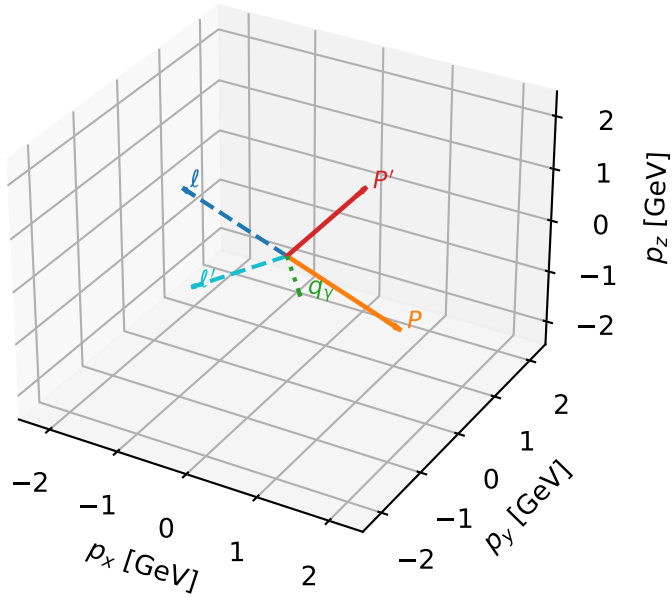


Leading initial-ensemble/final-state components (N=7, retained=98.2%)



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta

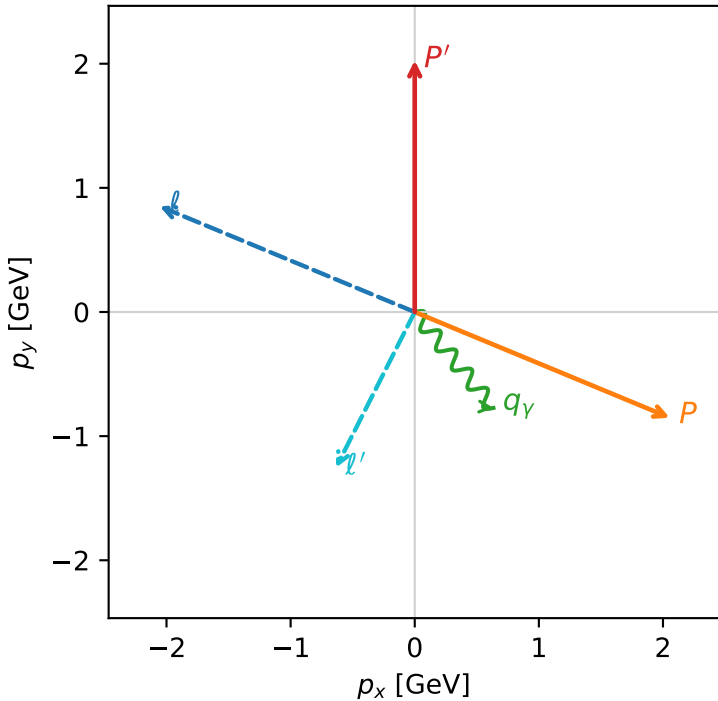


ghz_purity_mid_egamma_txtx_region_2 $P_{\text{GHZ}}=0.65415$
 region: mid_Egamma_TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

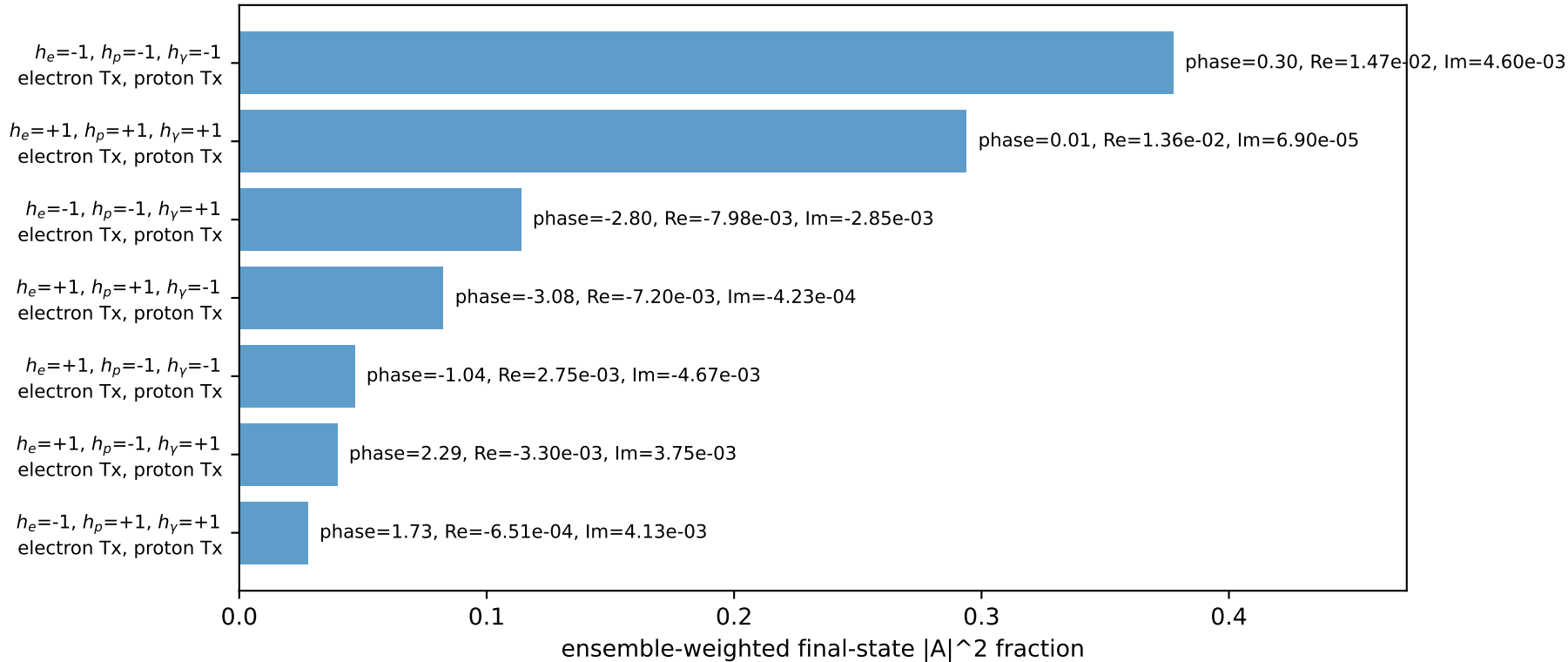
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.02682$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=2.74889$, $\phi_p=5.89049$
 $E_{\gamma}=1$, $\phi_{\gamma}=5.39961$
 $Q^2=5.77849$, $x_B=0.431917$, $t=-12.474$, $W^2=8.48005$, $y=0.648319$
 $\theta(\ell', \gamma)=66.2131$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -2.0551$ $p_y= 0.8512$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 2.0551$ $p_y= -0.8512$ $p_z= 0.0000$
 ℓ' $E= 1.4052$ $p_x= -0.6344$ $p_y= -1.2538$ $p_z= 0.0000$
 P' $E= 2.2333$ $p_x= 0.0000$ $p_y= 2.0268$ $p_z= 0.0000$
 q_{γ} $E= 1.0000$ $p_x= 0.6344$ $p_y= -0.7730$ $p_z= 0.0000$

Transverse plane

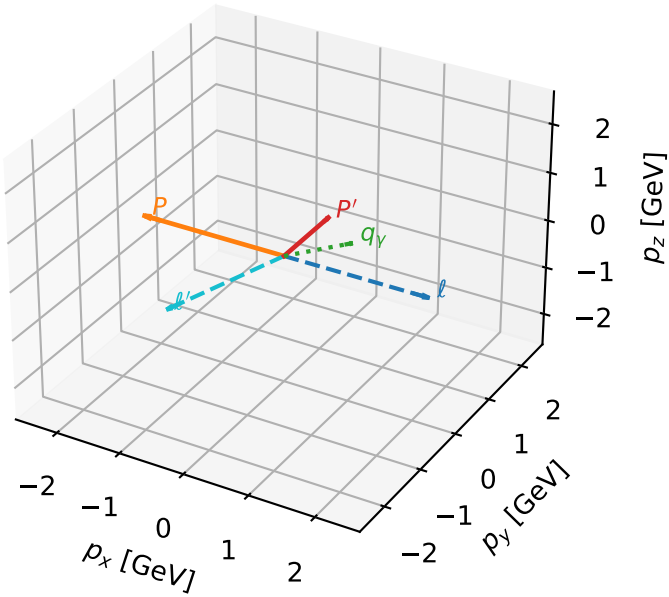


Leading initial-ensemble/final-state components (N=7, retained=98.2%)



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta



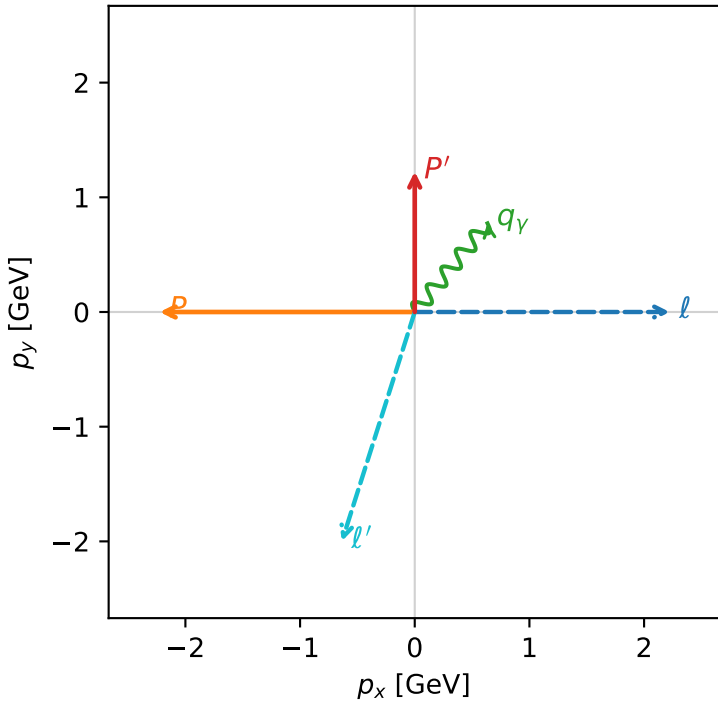
ghz_purity_mid_egamma_txtx_region_3 $P_{\text{GHZ}}=0.64149$
 region: mid_Egamma_TxTx_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.22462$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=0$, $\phi_p=3.14159$
 $E_\gamma=1$, $\phi_\gamma=0.883573$
 $Q^2=12.1468$, $x_B=0.910646$, $t=-5.68817$, $W^2=2.07171$, $y=0.646379$
 $\theta(\ell', \gamma)=158.243$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2244$ $p_x= 2.2244$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -2.2244$ $p_y= 0.0000$ $p_z= 0.0000$
 ℓ' $E= 2.0959$ $p_x= -0.6344$ $p_y= -1.9976$ $p_z= 0.0000$
 P' $E= 1.5426$ $p_x= 0.0000$ $p_y= 1.2246$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.6344$ $p_y= 0.7730$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=8, retained=100.0%)

